Lumina

Multi-Model Forecasting of Night-Time Light Data

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Motivation & Problem Statement

Why nighttime lights?

- **Urban growth planning** — anticipate electrified zone expansion
- **Energy demand estimation** — NTL correlates with electricity use
- **Disaster recovery** — NTL trajectory reveals recovery speed
- **Economic proxy** — tracks GDP in data-scarce regions

Kolkata NTL — 147 months of satellite data

The gap:

No publicly accessible platform for **multi-model NTL forecasting** of Indian cities with **live validation** against actual satellite

Dataset & Study Areas

VIIRS VCMSLCFG via Google Earth Engine | `avg_rad` band (nW/cm^2/sr) | 500 m/pixel | **147 monthly composites** (Jan 2014 – Mar 2026)

Kharagpur

Grid	35 × 45 pixels
Signal	Low, clear seasonality
Character	Campus city

Kolkata

Grid	70 × 70 pixels
Signal	High, complex heterogeneity
Character	Major metro (14M pop)

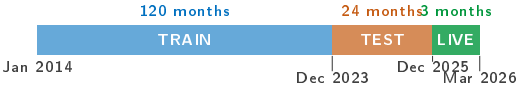
Annual seasonality and long-term upward trend visible in both cities

Preprocessing & Data Split

Preprocessing Pipeline

1. **Nodata masking** — pixels >1000 nW/cm^2/sr → NaN
2. **Mean brightness** — `np.nanmean` over valid pixels → scalar/month
3. **Outlier cleaning** — anomalous spikes replaced with local smooth value
4. **Frame stacking** (ConvLSTM only) — full spatial grids → `frames.npz`

Temporal Data Split



Key: Live validation data (Jan–Mar 2026) was downloaded **in April 2026**. Models were **completely frozen** before this data existed.

SARIMA — Statistical Baseline

What is SARIMA?

Classical Box-Jenkins model capturing trend, autocorrelation, and seasonality ($s = 12$ months) via differencing. Orders $(p, d, q)(P, D, Q)_{12}$ chosen by AIC grid search on training data.

Best Orders Found

City	Best Order	AIC
Kharagpur	$(0, 1, 1)(0, 1, 1)_{12}$	189.95
Kolkata	$(0, 1, 2)(0, 1, 1)_{12}$	444.22

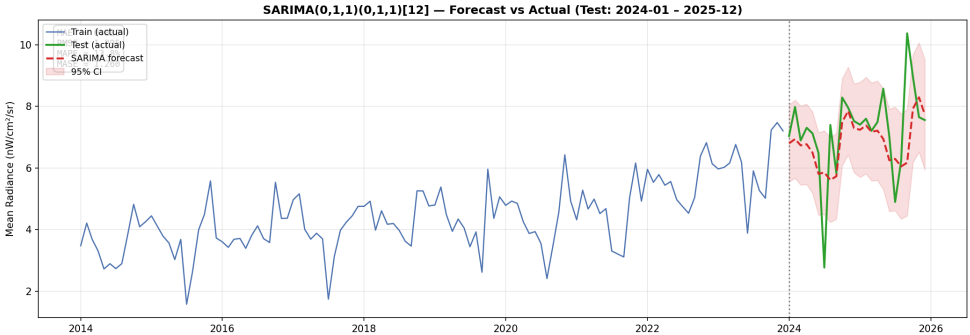
Test-set MAPE:

13.75% (Kharagpur) **24.82%** (Kolkata)

d=1, D=1: one regular + one seasonal difference removes both the upward trend and the annual cycle — a natural fit for monthly NTL data.

Key insight: Simple closed-form model outperforms deep networks on only 120 training observations — data scarcity matters.

SARIMA — Forecast vs Actual (Kharagpur)



SARIMA forecast vs actual — Kharagpur (Jan 2024 – Dec 2025) with 95% confidence interval

LSTM — Deep Learning Scalar Model

Architecture

```
Input (window × 1)
→ LSTM(units_1) + Dropout
→ LSTM(units_2) + Dropout
→ Dense(16, ReLU)
→ Dense(1)

Loss: MAE   Optimizer: Adam
```

MC Dropout: 50 stochastic forward passes at inference → mean = forecast, std dev = **uncertainty band**

Keras Tuner (30 trials)

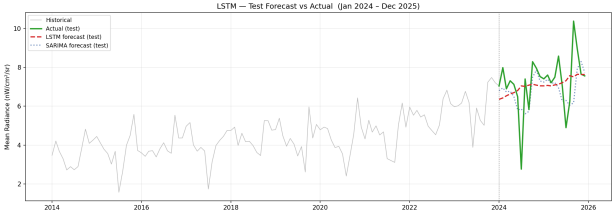
Param	Kgp	Kol
Window	12	24
LSTM-1	128	128
LSTM-2	32	16
Dropout	0.3	0.2
LR	0.0005	0.0005

Test-set MAPE:

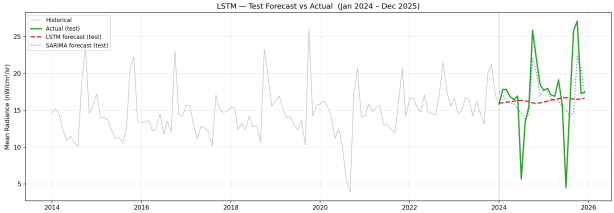
17.50% (Kharagpur) **30.50%** (Kolkata)

LSTM — Forecast vs Actual (with MC Dropout Uncertainty)

Kharagpur — Jan 2024 to Dec 2025



Kolkata — Jan 2024 to Dec 2025



ConvLSTM — Spatial Deep Learning Model

What makes it different?

- SARIMA & LSTM predict a **scalar** (mean brightness)
- ConvLSTM predicts a full **pixel-level spatial grid**
- Matrix multiplications in LSTM gates replaced by **2D convolutions**

Encoder-Decoder Architecture

```
Input: (T=12, H, W, 1)
Encoder: 3× ConvLSTM2D + BN
Bridge: Reshape → (1, H, W, 32)
Decoder: 2× ConvLSTM2D + BN
Output: Conv2D(1×1) → (H,W,1)
```

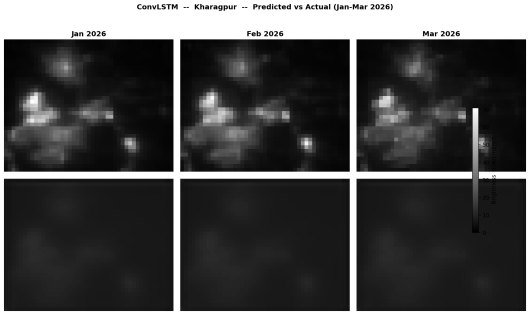
Spatial Metrics (Test Set)

Metric	Kgp	Kol
SSIM	0.477	0.536
PSNR (dB)	24.78	16.27
Pixel MAE	5.141	18.187

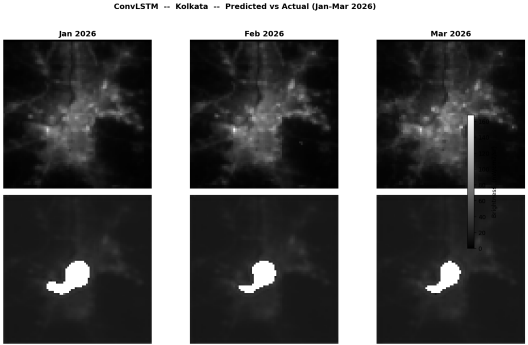
Unique value: Only model showing *which sub-regions* brighten or dim — invisible to SARIMA and LSTM.

ConvLSTM — Predicted vs Actual (Live Validation: Jan–Mar 2026)

Kharagpur



Kolkata



Grayscale brightness maps — brighter = higher NTL radiance. Models frozen before Jan–Mar 2026 data existed.

Comparative Results & Live Validation ★

All models validated against **real Jan–Mar 2026 VIIRS observations** downloaded in April 2026 — models were completely frozen before this data existed.

Kharagpur (nW/cm^2/sr)

Month	Actual	SARIMA	err%	LSTM	err%
Jan 2026	9.154	8.182	10.6	7.617	16.8
Feb 2026	9.462	8.486	10.3	7.390	21.9
Mar 2026	8.869	8.058	9.1	7.648	13.8
Mean			**10.0%**		17.5%

Month	ConvLSTM	err%
Jan 2026	7.743	15.4
Feb 2026	7.738	18.2
Mar 2026	7.726	12.9

Kolkata (nW/cm^2/sr)

Month	Actual	SARIMA	err%	LSTM	err%
Jan 2026	20.823	18.132	12.9	16.467	20.9
Feb 2026	21.537	18.345	14.8	16.628	22.8
Mar 2026	22.328	17.654	20.9	16.840	24.6
Mean			**16.2%**		22.8%

Month	ConvLSTM	err%
Jan 2026	29.401	43.0
Feb 2026	27.331	28.5
Mar 2026	26.034	18.1

Live Demo — Lumina Dashboard

lumina-iamabhinav-dev.streamlit.app

Stop 1 — NTL Explorer	drag time slider Jan 2014 → Jan 2026
Stop 2 — Change Detection	Jan 2014 vs Jan 2026 signed difference map
Stop 3 — 2026 Validation	all three model lines vs actual — SARIMA wins live
Stop 4 — ConvLSTM Spatial	predicted vs actual frame grid + error heatmap

8-page interactive app	Publicly deployed	Open source
Streamlit Community Cloud	github.com/iamabhinav-dev/Lumina	Auto-deploys on push

Conclusions

1. **SARIMA is the best scalar forecaster** — **10.0%** live error (Kharagpur), **16.2%** (Kolkata). Explicit s=12 seasonal structure is a natural fit for monthly NTL.
2. **LSTM provides probabilistic forecasts** — competitive deep-learning baseline with principled uncertainty quantification via Monte Carlo Dropout.
3. **ConvLSTM adds spatial intelligence** — pixel-level maps of city brightness changes, information invisible to scalar models (SSIM 0.477 / 0.536).
4. **All validated against real 2026 satellite data** — models frozen before live observations were downloaded, confirming reliability beyond the training distribution.
5. **Lumina dashboard** publicly deployed — end-to-end platform from raw GEE export to interactive forecasting visualisations.

Future Work

Near-Term

- Extend to Mumbai, Delhi, Bengaluru, Chennai — pipeline already city-parameterised via `src/cities.py`
- Longer live validation as 2026 composites accumulate (Apr – Dec 2026)
- Add **Temporal Fusion Transformer** as a 4th model
- Attention gates + global pre-training for ConvLSTM on Kolkata

Research Paper Proposal

"Multi-Modal Night-Time Light Analytics: VIIRS + WorldPop Population + District GDP for Urban Growth Prediction in Indian Cities"

Key additions:

- WorldPop gridded population data
- District-level NSDP economic statistics
- Multi-input LSTM / Transformer fusion
- NTL – GDP lead-lag correlation study across 10+ cities

Thank You

Dashboard:	lumina-iamabhinav-dev.streamlit.app
GitHub:	github.com/iamabhinav-dev/Lumina
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Questions?